
You use the apps. What would you build?

Careers in AI engineering and full-stack development.

Show of hands: who used an AI tool this week? What did you use it for?

AUDIENCE

Classes IX to XII

VENUE

KV RK Puram, Sector 2

DEPARTMENT

DEPFE, NCERT, New Delhi

Tushar Gaurav

Full Stack Software Engineer

Software Developer at Orangewood Labs
Noida, India

tushgaurav.com/resume

B.Tech in Computer Science Engineering
Noida Institute of Engineering and Technology, 2020 to 2024

RoboGPT

Web applications that make robotic arm programming easier for non-technical users.

Robot-control interfaces

Real-time 3D visualisation and connected systems using gRPC and WebSockets.

From software to deployment

Python, TypeScript, React, Next.js and cloud infrastructure for real client projects.

You tap "Order". What happens next?

Name one thing the app must do before your food arrives.

The screen

Menus, buttons and checkout.

Frontend is what the user sees and interacts with.

The logic

Validate the order and process requests.

Backend is what happens behind the scenes.

The data

Store users, items and orders.

Databases, servers and APIs connect the whole application.

A full-stack developer builds and maintains both sides of a web application.

Would you enjoy **fixing what broke?**

What you do

- Build and maintain websites and web applications
- Design interfaces and write server-side logic
- Connect databases, test and debug issues
- Deploy to servers and solve problems with a team

What the day is like

- Usually **8 to 9 hours a day, 5 days a week**
- Office or remote work; timing may be flexible
- Extra hours around project deadlines
- Desk-based work, screen time and mental fatigue: breaks and eye care matter

What helps you succeed

- Logical reasoning and basic to intermediate maths
- Curiosity, creativity and attention to detail
- Patience, persistence and continuous learning
- Communication, teamwork, time management and reliability under pressure

Quick check: which sounds most like you: designing, solving puzzles or debugging?

What can your **first step** look like?

Education and eligibility

- **10+2, any stream**, with Mathematics preferred
- The sheet lists **50% minimum**; actual marks and entrance requirements depend on the college
- B.Tech / B.E. in Computer Science, IT or Software Engineering; or B.Sc in Computer Science / IT; or BCA
- Alternative: **BCA (3 years), then MCA (2 years)**

Where you can work

- IT and product companies, startups and unicorns
- Freelance, remote work or your own product / agency
- Government and public sector, including NIC and BSNL
- E-commerce, banking, healthcare, edtech, gaming and entertainment

Entry / 0 to 2 years

₹3 to 6 LPA

Mid level / 3 to 5 years

₹6 to 12 LPA

Senior / 6+ years

₹12 to 25+ LPA

Indicative annual India salary ranges from the supplied sheet, not guarantees. Pay varies by skills, employer and location.

Leave with a shortlist. **Not just a job title.**

Delhi institutions in the sheet

- IIT Delhi, NSUT and DTU
- IIIT Delhi and Jamia Millia Islamia
- MAIT, MSIT and BPIT
- Check each institution's current courses, eligibility and admission rules

Courses, practice and internships

- Coursera and Udemy
- LinkedIn Learning
- Internshala for internships
- Stack Overflow for programming questions

Career information and outlook

- NCERT career cards / PSSCIVE
- NIRF rankings and National Career Service
- The sheet highlights AI, cloud, digital transformation and startup opportunities
- Freelancing and remote work also feature in the outlook

Pick one college to research, one course to try and one problem to build for.

What does a career in AI really involve?

A structured walk through everything you need to decide whether this profession is right for you.

01 Educational Pathway

Two routes: degree or self taught

02 Nature of Work

Day to day tasks and tools

03 Working Conditions and Attributes

Environment and skills that help

04 Career Progression and Salary

Growth from entry to leadership

05 Opportunities and Colleges

Where to work and where to study

06 Courses and Future Outlook

Building employability for the long run

What can you build before you graduate?

A degree can give you structure. A working project gives you something to demonstrate. You can build both at the same time.

A study planner?
A club website?
A lost-and-found app?

Choose a small problem you understand. Who would use your solution?

Two ways into the field

There is no single fixed route. Pick the one that fits you, or blend both.

ROUTE A

The degree route

Structured, credential first

- 1 Science with Maths in Class XI and XII**
Add Computer Science if offered, and start Python early.
- 2 Crack an entrance exam**
JEE, BITSAT or CUET for IITs, NITs, IIITs and BITS.
- 3 B.Tech or B.Sc in CSE, AI-ML or Data Science**
Optional M.Tech or PhD for research and senior roles.
- 4 Internships, then a full time job**
Campus placements and referrals open the first role.

ROUTE B

The **builder** route

Self taught, portfolio first

- 1 Learn one language deeply**
Master Python and the maths behind machine learning.
- 2 Build real projects**
Ship them on GitHub and let the work speak for you.
- 3 Find clients and freelance**
Take paid gigs, small contracts and open source work.
- 4 Improve, specialise, grow**
Turn a real track record into full time and senior roles.

The AI sounds confident. **Is it correct?**

Would you trust an answer without checking it? An AI engineer tests more than the demo.

Core Responsibilities

- Design, train and fine tune **machine learning and deep learning** models
- Collect, clean and prepare data, then engineer features
- Deploy models into products and **monitor them in production**
- Work on NLP, computer vision, recommendations and large language models
- Write reliable, production grade code and run experiments

Tools and Technologies

- **Languages.** Python, SQL, sometimes C++ or Java
- **Frameworks.** PyTorch, TensorFlow, scikit learn, Hugging Face
- **Cloud and DevOps.** AWS, GCP, Azure, Docker, Git
- **Data.** Pandas, NumPy, Spark, vector databases
- Collaborates with data scientists, product and software teams

The environment and what helps you succeed

Hours and Setup

- Typical **40 to 45 hour** work weeks
- Extra hours near launches or deadlines
- **Hybrid and remote** work are common
- On site work for robotics and hardware

Teams and Culture

- Small **cross functional teams**
- Agile sprints, code reviews, stand ups
- Global collaboration across time zones
- Constant need to keep learning

Personal Attributes

- Strong **mathematics** and logical reasoning
- **Curiosity** and self driven learning
- Patience for experimentation and debugging
- Explaining complex ideas **simply**

Same job title. Different responsibility.

A career grows as you move from completing tasks to owning outcomes. Years of experience alone do not tell the whole story.

Who would you hire?
The person with ten
certificates, or a project
real people use?

What else would you want to know before
deciding?

From entry level to leadership

Intern or Trainee

Starting out

Learn on real projects, assist with data and experiments while completing studies.

Junior AI or ML Engineer

0 to 2 years

Build and test models under guidance. Strengthen coding and data skills.

AI or ML Engineer

2 to 5 years

Own features end to end, deploy models to production, mentor juniors.

Senior AI Engineer

5 to 8 years

Lead complex systems, set technical direction, review architecture.

Lead, Principal or ML Architect

8 to 12 years

Design large scale AI systems and guide multiple teams.

Head of AI, Director or CTO

12 plus years

Own AI strategy and organisation wide impact.

Indicative compensation in India

Entry or Junior

0 to 2 years

First roles. Varies widely by employer and city.

6 to 12 LPA

Mid level

2 to 5 years

Proven delivery and production experience.

12 to 25 LPA

Senior

5 to 8 years

Technical ownership and mentoring.

25 to 45 LPA

Lead or Principal

8 to 12 years

Architecture and multi team leadership.

45 to 80 LPA

Director or Head of AI

12 plus years

Strategy. Top product firms pay well above.

80 LPA plus

Figures are broad, indicative annual ranges in Indian Rupees and vary significantly with company, location, skills and global roles. Top product companies and overseas positions can pay considerably more.

Where the demand is

Sectors

- IT and software, e commerce
- Fintech, healthcare, automotive
- Defence, agriculture, telecom
- Edtech, media and entertainment

Organisations

- Global. Google, Microsoft, Amazon, NVIDIA, Meta
- Indian IT. TCS, Infosys, Wipro
- Startups. Sarvam AI, Krutrim and more
- Research. ISRO, DRDO, IITs, IISc

Geographies

- Bengaluru, Hyderabad, Pune
- Gurugram, NCR, Chennai
- Global. US, Europe, Singapore, UAE
- Remote, freelance and consulting

Recommended institutions

Premier Engineering Institutes

- **IITs.** Bombay, Delhi, Madras, Kanpur, Kharagpur, Hyderabad, which has a dedicated AI department
- **IISc Bangalore** for leading AI research
- **NITs.** Trichy, Surathkal, Warangal
- **BITS Pilani.** Pilani, Goa, Hyderabad

Strong CS and AI Programmes

- **IIITs.** Hyderabad, Delhi, Bangalore
- Delhi University, DTU and NSUT
- VIT, Manipal, Ashoka University
- Reputed state and private universities with AI or DS courses

Building employability for the long run

Courses and Certifications

- DeepLearning.AI specialisations by Andrew Ng
- Google, AWS and Azure ML certifications
- Stanford CS231n for vision, CS224n for NLP
- **Kaggle** competitions and Kaggle Learn

Internships and Portfolio

- Research internships at IITs and IISc
- Google Summer of Code and industry internships
- Open source and a **GitHub project portfolio**
- Build things and share them publicly

Future Outlook

- **Generative AI and agents** are reshaping every industry
- Government push through the **IndiaAI Mission**
- Rising focus on **responsible and ethical AI**
- One of the fastest growing, highest demand careers

YOUR NEXT SEVEN DAYS

One problem. One small working project.

Pick a problem at school or home. Build the smallest useful version. Show it to one person and improve it from their feedback.

Tell us: what will you build, and who is it for?

Questions? Ask about getting started, college, clients or the work itself.